

Find first non-repeating character in a string.

```
package String;
import java.util.HashMap;
import java.util.Map;
public class First_non_repeat_character {
    public static void main(String[] args) {
        String input = "swiss";
        char result = nonrepeat(input);
        if (result != '\0') {
            System.out.println("The first non-repeat character: " + result);
        } else {
            System.out.println("All characters repeat.");
        }
    }

    public static char nonrepeat(String s) {
        Map<Character, Integer> counts = new HashMap<>();
        for (char c : s.toCharArray()) {
            counts.put(c, counts.getOrDefault(c, 0) + 1);
        }
        for (char c : s.toCharArray()) {
            if (counts.get(c) == 1) {
                return c;
            }
        }
        return '\0';
    }
}
```

The first non-repeat character: w

Remove duplicate characters from string.

```
package String;
import java.util.LinkedHashSet;
import java.util.Set;
public class Remove_duplicate {
    public static void main(String[] args) {
        String st="AaBbCCDDDeF";
        String s=st.toLowerCase();

        Set<Character> set = new LinkedHashSet<>();
        for (char c : s.toCharArray()) {
            set.add(c);
        }
        StringBuilder res = new StringBuilder();
        for (char c : set) {
            res.append(c);
        }
        System.out.println("Original: " + st);
        System.out.println("Result: " + res.toString());
    }
}
```

```
}  
Original: AaBbCCDDeF  
Result: abcdef
```

Find element with maximum frequency.

```
package String;  
import java.util.*;  
public class Max_freq {  
    public static void main(String[] args) {  
        String s="aaabjbfjfodcfjjjjjfnert";  
  
        Map<Character, Integer> counts=new HashMap<>();  
        for (char c : s.toCharArray()) {  
            counts.put(c, counts.getOrDefault(c, 0) + 1);  
        }  
  
        char maxChar = ' ';  
        int maxCount = 0;  
        for (Map.Entry<Character, Integer> entry : counts.entrySet()) {  
            if (entry.getValue() > maxCount) {  
                maxCount = entry.getValue();  
                maxChar = entry.getKey();  
            }  
        }  
        System.out.println("Character of max frequency: " + maxChar);  
        System.out.println("Frequency: " + maxCount);  
    }  
}  
Character of max frequency: j  
Frequency: 8
```

Sort a map by keys.

```
package Map_pract;  
import java.util.*;  
public class Sort_map_keys {  
    public static void main(String[] args) {  
  
        Map<String, Integer> map = new HashMap<>();  
        map.put("Zebra", 10);  
        map.put("Apple", 50);  
        map.put("Mango", 30);  
        TreeMap<String, Integer> ms = new TreeMap<>(map);  
        System.out.println("Sorted: " + ms);  
    }  
}  
Sorted: {Apple=50, Mango=30, Zebra=10}
```